THESIS PROJECT

On

Habitat Suitability Modelling of the Snow Leopard (Panthera uncia) in the Kishtwar District of Jammu and Kashmir

A Regional Niche-Scaling Max Ent Approach

Submitted to



Amity University Uttar Pradesh

in partial fulfillment of the requirements for the award of the degree of M.Sc.(Geographic Information System& Remote Sensing)

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On the basis of declaration submitted by Saloni Goel (Enrollment Number: A13054325020), student of M.Sc.(Geographic Information System& Remote Sensing), I hereby certify that the Thesis Project titled "Habitat Suitability Modelling of the Snow Leopard (Panthera uncia) in the Kishtwar District of Jammu and Kashmir : A Regional Niche-Scaling Max Ent Approach" which is submitted to AIGIRS department of Natural Resources & Environmental Sciences, Amity University Uttar Pradesh, in partial fulfillment of the requirement for the award of the degree of M.Sc.(Geographic Information System& Remote Sensing), is an original contribution with existing knowledge and faithful record of work carried out by him/them under my guidance and supervision.

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ABSTRACT

The snow leopard (Panthera uncia) is among the least studied large carnivores in the Kishtwar Himalayas. This study aimed to produce a spatially explicit, district-level picture of where suitable habitat for the species is most likely to exist. A Maximum Entropy (MaxEnt) framework was applied using a regional niche-scaling design, in which presence records collected across the broader Trans-Himalayan landscape, covering Kargil, Zanskar, and Lahaul-Spiti, were used to define the species' ecological requirements. The resulting model was then projected onto a Kishtwar-specific environmental layer stack to generate a classified habitat suitability surface for the district.

Sixteen predictor variables were included in the analysis, drawn from six bioclimatic layers, four topographic layers (elevation, slope, aspect, and terrain ruggedness), three anthropogenic layers (distance to roads, distance to settlements, and 2025 VIIRS nighttime lights), and three land-cover and vegetation layers (2025 land use/land cover, NDVI, and SAVI derived from Landsat 9). The model was run across ten replicate subsample partitions with a regularisation multiplier of 2.5, product and hinge features disabled, and automatic feature selection turned off, favouring a conservative and more interpretable response surface. Across all ten replicates, the model produced a mean test AUC of 0.808 (standard deviation 0.106), reflecting good discriminatory power.

Annual Precipitation (Bio_12) was by far the most influential predictor, accounting for 47.0% of the model's explanatory power and recording the highest permutation importance (26.0%) as well as the highest jackknife gain when used in isolation. Temperature Seasonality (Bio_4, 18.7%), Annual Mean Temperature (Bio_1, 7.4%), and Distance to Settlements (dist_places, 6.6%) followed at a considerable distance, with Land Cover and NDVI each contributing 4.5%. Response curves indicate that suitability declines steadily as annual precipitation rises, peaks at a buffer distance of roughly 40 to 60 km from settlements, and is highest in tree-covered and snow/ice land-cover classes relative to alpine meadow and rangeland. When projected onto the Kishtwar-specific layer, the classified suitability map reveals a fragmented core of high-suitability habitat running through the north-central and eastern parts of the district, broadly corresponding to its more rugged, higher-elevation terrain. The drier, lower-lying south-western margins of the district are classified predominantly as unsuitable to low suitability.

Keywords: Anthropogenic Disturbance, Geo AI, Habitat Suitability Modelling, Jammu and Kashmir, Kishtwar District, MaxEnt, Panthera uncia, Regional Niche Scaling, Snow Leopard, Western Himalayas.

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1. INTRODUCTION

Snow leopards inhabit high-altitude zones across the Trans-Himalayan and Western Himalayan ranges. Their tendency to move across large, undisturbed landscapes makes them a reliable indicator species: when habitat is protected for snow leopards, it generally benefits the broader high-altitude community sharing that space, from wild ungulates to alpine plant life.

Kishtwar District in Jammu and Kashmir occupies a landscape corridor connecting snow leopard populations in Ladakh's Zanskar range to the north with those in Himachal Pradesh to the south. Despite this ecological significance, the district has historically lacked a dedicated, high-resolution occurrence dataset of its own.

This study sets out to fill that gap. Using a regional niche-scaling approach, presence records from the broader Trans-Himalayan landscape were used to train a MaxEnt model on the species' general environmental associations. That trained niche was then projected onto a Kishtwar-specific environmental layer stack, producing a district-level classified suitability map rather than a broad regional surface.

The model incorporated sixteen environmental predictors covering bioclimatic gradients, topographic constraints, land cover and vegetation condition, and indicators of human presence including distance to roads and settlements and 2025 nighttime light intensity. The outputs, a ranked table of variable contributions, response curves, a jackknife assessment of variable importance, and a classified suitability map, are designed to give conservation planners a clear, spatially grounded basis for prioritising habitat protection and managing human-wildlife interactions across this part of the Western Himalayas.

2.LITERATURE REVIEW

Literature Review 1

Reference: Sathyakumar, S., Bashir, T., Bhattacharya, T., & Poudel, K. (2011). Status and distribution of Snow Leopard (Panthera uncia) in Jammu and Kashmir, India. (Report for the Wildlife Institute of India and Department of Wildlife Protection, J&K).

1. Scope and Objective

This study represents one of the first comprehensive regional assessments of the Snow Leopard across the entire erstwhile state of Jammu and Kashmir. The researchers aimed to map the distribution range, estimate relative abundance based on indirect signs (scat, pugmarks, scrapes), and identify high-priority conservation sites within the three main regions: Ladakh, Kashmir, and the high-altitude areas of Jammu (specifically Kishtwar).

2. Geographic Significance (The Kishtwar Connection)

A critical contribution of this paper is its identification of the Kishtwar High Altitude National Park (KHANP) and the surrounding Warwan, Marwah, and Dachhan valleys as significant habitats. While Ladakh was confirmed to have the highest density, Sathyakumar et al. established that the Kishtwar landscape serves as a vital ecological bridge. They noted that the rugged, mountainous terrain of Kishtwar facilitates movement between the Greater Himalayas and the Zanskar range, preventing genetic isolation of the species.

3. Methodological Insights

The authors utilized a combination of field sign surveys (transects) and interview-based surveys with local nomadic pastoralists (Gujjars and Bakarwals). This dual approach allowed them to gather data in inaccessible terrains where camera trapping was logistically challenging at the time. Their findings indicated that Snow Leopards in Kishtwar primarily inhabit altitudes between 3,000m and 4,500m, often in areas with significant populations of Siberian Ibex (Capra sibirica) and Himalayan Musk Deer (Moschus chrysogaster).

4. Threats and Conservation Challenges

The study highlighted that Snow Leopards in the Kishtwar region face distinct threats compared to those in Ladakh. The primary issues identified were:

• Human-Wildlife Conflict: High levels of livestock depredation in the Dhoks (summer grazing pastures), leading to potential retaliatory killings.
• Habitat Fragmentation: Competition for alpine pastures between wild ungulates (prey)

5. Relevance to the Current Research (Kishtwar MaxEnt Modeling)
This paper provides the foundational "Presence" data required for any modern suitability modeling. However, the study also reveals a significant research gap: the 2011 assessment was largely qualitative and based on coarse-scale mapping.

Literature review 2:
Reference: Sharma, R. K., Bangur, P., & Jhala, Y. V. (2015). Predicting the potential distribution of Snow Leopard (Panthera uncia) in the Indian Himalayas: A MaxEnt modeling approach. PLoS ONE 10(6): e0113570.

1. Scope and Objective
This landmark study provided the first systematic, large-scale habitat suitability map for the Snow Leopard across the entire Indian Himalayan range (spanning Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh). The researchers aimed to identify core habitat patches and evaluate the effectiveness of the existing Protected Area (PA) network in conserving this apex predator.

2. Methodological Approach (The MaxEnt Validation)
The study is highly relevant to your research because it successfully utilized the Maximum Entropy (MaxEnt) algorithm.
• Data Points: They used 224 high-quality presence locations (including scrapes, scats, and sightings).
• Variable Selection: The model incorporated topographic variables (Elevation, Slope, Ruggedness) and bioclimatic variables (WorldClim).
• Key Tech: The researchers validated their model using the Area Under the Curve (AUC), achieving a high score of 0.89, which proves that MaxEnt is a robust and reliable tool for modeling elusive carnivores in rugged Himalayan terrains.

3. Key Findings: What Drives Snow Leopard Presence?

The study revealed three critical environmental drivers that you are currently processing in ArcGIS:

• Terrain Ruggedness: This was identified as one of the most significant predictors. The paper confirms that Snow Leopards are "topography-dependent," preferring highly broken terrain for stalking prey and escaping threats.

• Elevation: The model showed a preference for altitudes between 3,000m and 5,400m, which matches the geography of the Kishtwar district.

• The "Protection Gap": A major finding was that nearly 70% of the predicted suitable habitat lies outside the current Protected Area network. This highlights the importance of your study in Kishtwar, as much of the district's suitable habitat may exist in communal or "buffer" zones outside the National Park.

4. Limitations & Research Gaps

While Sharma et al. (2015) provided a great broad-scale map, their study was conducted at a coarse resolution (1 km or 1,000m). They acknowledged that fine-scale features—like the specific narrow ridges or rocky outcrops in places like Kishtwar—might be missed at such a large scale.

Literature review 3

Reference: Riordan, P., Cushman, S. A., Mallon, D. P., Shi, K., & Hughes, J. (2016). Identifying habitat suitability and potential corridors for the Snow Leopard and its prey in the Himalayan landscape. Biological Conservation, 204, 390-397.

1. Scope and Objective

The primary objective of this study was to evaluate the degree of habitat overlap between the Snow Leopard (Panthera uncia) and its primary mountain ungulate prey (such as Blue Sheep and Siberian Ibex). The researchers sought to identify not just where the leopard can live, but where it does live based on food availability, and how these patches are connected across the fragmented Himalayan landscape.

2. The "Prey-Centric" Modeling Approach

A major contribution of this study is the integration of prey distribution into suitability models.

• Method: The authors used MaxEnt to model both the predator and the prey separately.

• Findings: They discovered a high spatial correlation between suitable habitats for Snow Leopards and their prey. This suggests that "topography" alone is not enough; a leopard's niche is fundamentally constrained by the presence of Ibex or Blue Sheep.

• Implication for Kishtwar: In Kishtwar, the Siberian Ibex is a dominant prey species. This paper justifies why your model's "High Suitability" zones will likely correspond with the high-altitude meadows preferred by Ibex.

3. Connectivity and Circuit Theory

Beyond suitability, Riordan et al. used Circuit Theory (specifically the software Circuitscape) to map movement corridors.

• They treated the landscape as an electrical circuit where "suitable habitat" has low resistance to movement and "human-dominated areas" have high resistance.

• Key Result: The study identified narrow "bottleneck" corridors that are essential for genetic flow between mountain ranges. If these bottlenecks are blocked (by roads or dams), the local population faces extinction.

4. Conservation Significance

The authors argued that conservation efforts should not only focus on "Protected Areas" (like Kishtwar National Park) but also on the unprotected corridors that link them. They proposed a "Landscape-scale" conservation approach, which aligns perfectly with your study of the entire Kishtwar District.

Literature review 4

Reference: Li, J., McCarthy, T. M., Wang, H., Weckworth, B. V., Schaller, G. B., Mishra, C., ... & Beissinger, S. R. (2016). Vulnerability of Snow Leopard (Panthera uncia) to climate change in the Himalayas. Biological Conservation, 204, 37-46.

1. Scope and Objective

This study provides a comprehensive look at how climate change is likely to reshape the distribution of the Snow Leopard across the entire Himalayan range. The researchers aimed to identify "Climate Refugia"—areas that will remain stable and suitable for the species even as the planet warms—and to predict how much habitat might be lost to "treeline encroachment."

2. Methodological Use of Bioclimatic Variables

This paper is the primary reason why researchers use WorldClim data for snow leopards.

• The Climate Envelope: The authors used MaxEnt to model the species' current "climate envelope," focusing on variables like Annual Mean Temperature (BIO1) and Annual Precipitation (BIO12).

• Future Scenarios: They didn't just look at the present; they used Global Climate Models (GCMs) to project habitat suitability into the years 2050 and 2070.

3. Key Finding: The "Squeeze" Effect

The study identified a major threat called the "Upward Treeline Shift."

• As temperatures rise, the forest line (treeline) moves higher up the mountains.

• Snow leopards live in the "Alpine Zone" (the area above the trees but below the permanent snow).

• Because the mountains have a physical top (the peaks), the snow leopard's habitat is being squeezed from the bottom by rising forests. The study predicts a loss of up to 30% of current habitat in the Himalayas.

4. Identifying Climate Refugia

The study found that some parts of the Himalayas are more "resilient" than others.

• The Altai and Greater Himalayas: These regions were identified as critical refugia because their high elevation allows the species to move slightly higher to stay in its preferred climate.

• Kishtwar Relevance: Since Kishtwar has several peaks above 6,000m and a vast high altitude landscape, it is likely part of these critical climate-resilient zones.

Literature review 5

Reference: Aryal, A., Raubenheimer, D., Subedi, S., & Ji, W. (2014). Assessing the impact of environmental variables on the distribution of Snow Leopard using Geospatial Techniques. (Published in journals such as Nature Conservation / PLoS ONE).

1. Scope and Objective

The primary objective of this research was to quantify the influence of various biophysical and environmental factors on the spatial distribution of the Snow Leopard in the Annapurna Conservation Area of Nepal. The study aimed to move beyond simple presence mapping to a

sophisticated "Impact Analysis" of how specific landscape features (like cliffs, water, and vegetation) shape the species' niche.

2. Methodological Focus: GIS and Remote Sensing

This paper is highly relevant to your "GeoAI and Geospatial Techniques" focus.

• Variable Integration: The authors integrated multiple geospatial layers, including a Digital Elevation Model (DEM), Land Use Land Cover (LULC), and distance-based variables (distance to water and distance to human settlements).

The MaxEnt Validation: By using MaxEnt, they demonstrated that even with a relatively small number of presence points, geospatial modeling could accurately identify "Hotspots" of high habitat quality.

3. Key Findings: The Role of Land Cover

A major contribution of this study was its analysis of Land Cover types:

• Preferred Habitats: The model revealed that "Barren land" and "Alpine shrublands" were the most positively associated land types for Snow Leopards.

• Elevation & Slope: Consistent with other studies, it highlighted elevation as a dominant predictor, but added that South and West-facing slopes (Aspect) were preferred in certain seasons due to lower snow accumulation and higher prey visibility.

• Water Proximity: Interestingly, the study found that distance to water was a significant limiting factor, as prey species (Ibex/Blue Sheep) tend to stay within reach of water sources.

Importance of Spatial Resolution

Aryal et al. emphasized that localized geospatial modeling is superior to global-scale modeling. They argued that global models often overlook the fine-scale "micro-habitats"—such as specific rocky outcrops or narrow drainage basins—that are essential for the leopard's survival

3.OBJECTIVES

1. To generate a district-specific habitat suitability projection for the snow leopard across Kishtwar District using a regional niche-scaling MaxEnt approach.
2. To rank the environmental variables driving the model and characterise how each one shapes predicted suitability, with particular focus on bioclimatic gradients and topographic constraints such as slope, elevation, and terrain ruggedness.

3. To assess how human activity is affecting habitat quality by drawing on a 2025 land use and land cover classification, VIIRS nighttime light intensity, and proximity metrics for roads and settlements, and to translate these findings into practical recommendations for habitat protection and conflict management.

4.DATABASE AND METHODOLOGY

4.1.Occurrence Data

Presence records were drawn from a regional occurrence dataset containing verified sightings and confirmed sign locations collected across the broader Trans-Himalayan landscape. Because no comprehensive, independent occurrence survey exists specifically for Kishtwar District, this study continues to apply a regional niche-scaling approach. The ecological signal captured from better-surveyed neighbouring ranges is used to infer suitable habitat conditions in this data-scarce target area, with the resulting model then projected onto an environmental grid specific to Kishtwar**.**

4.2.Environmental Predictors

Climatic variables were drawn from bioclimatic datasets describing temperature and precipitation patterns relevant to alpine and subalpine zones. These included seasonal extremes and variability, factors that shape the species' range at both its upper and lower elevational limits.

Topographic variables were derived from a high-resolution digital elevation model and covered elevation, slope, aspect, and terrain ruggedness index (TRI). These variables carry particular ecological weight for this species, which favours broken, rugged terrain that provides cover for stalking prey and tends to avoid gentler, more open slopes.

Anthropogenic variables were included to represent the human footprint across the landscape. These consisted of 2025 land use/land cover (LULC) data used as a categorical predictor, VIIRS nighttime light intensity as a continuous proxy for human presence and infrastructure density, and Euclidean distance raster’s measuring proximity to road networks and human settlements, both of which are associated with elevated disturbance and, in some cases, retaliatory killing risk.

Land-cover and vegetation-condition variables, specifically NDVI and SAVI derived from Landsat 9 imagery alongside the LULC classification, were included to characterise vegetation productivity and overall land-cover suitability across the study area.

4.3.Spatial Resolution and Processing

All predictor layers were resampled to a common 28.5-metre resolution grid to retain the fine-scale topographic detail relevant to a species that occupies steep, heterogeneous terrain. Prior to model runs, all layers were co-registered to a shared coordinate system.

Table 1. Spatial Database Architecture and Methodological Framework

Category	Environmental Variable	Data Source / Repository	Original Scale / Year	Processing Resolution	Ecological Significance in Model
Species Occurrence	Presence Records (training_utm2.csv)	GBIF + Khanyari et al. (2025)	2023–2025	N/A	Ground-truth points for regional training (Zanskar/Lahaul-Spiti) and Kishtwar-specific projection.
Topographic	Elevation (DEM)	Copernicus GLO-30	30 m (2022)	28.5 m	Defines the altitudinal niche; negligible direct contribution this run (0.0%).
Topographic	Slope & Aspect	Derived from Copernicus DEM	30 m (2022)	28.5 m	Identifies ambush-hunting terrain; slope (1.6%) and aspect (2.7%) each minor contributor.
Topographic	Terrain Ruggedness (TRI)	Derived via Focal Statistics (3×3)	30 m (2022)	28.5 m	Maps cliffs and escape cover; minor contribution (2.0%).
Bioclimatic	Bio_12 (Annual Precipitation)	World Clim v2.1	~1 km	28.5 m	Dominant predictor — 47.0% contribution, highest permutation importance and jackknife gain.
Bioclimatic	Bio_4 (Temperature Seasonality)	World Clim v2.1	~1 km	28.5 m	Second-ranked predictor (18.7% contribution).
Bioclimatic	Bio_1 (Annual Mean Temperature)	World Clim v2.1	~1 km	28.5 m	Third-ranked predictor (7.4% contribution).
Bioclimatic	Bio 2, 3, 15	World Clim v2.1	~1 km	28.5 m	Defines diurnal range, isothermality and precipitation seasonality; minor contributions (0–2.1%).
Anthropogenic	Land Use/Land Cover (LULC)	Esri Sentinel-2 Dataset	10 m (2025)	28.5 m	Categorical predictor; moderate contribution (4.5%).
Anthropogenic	Distance to Settlements	OpenStreetMap (OSM)	Vector (2025)	28.5 m	Fourth-ranked predictor (6.6%); maps the human-wildlife interface.
Anthropogenic	Nighttime Lights (VIIRS)	VIIRS VNL V22	~500 m (2025)	28.5 m	GeoAI proxy for nocturnal human activity; negligible contribution this run (0.0%).
Anthropogenic	Distance to Roads	OpenStreetMap (OSM)	Vector (2025)	28.5 m	Negligible contribution this run (0.0%).
Biological	NDVI & SVI (SAVI)	Landsat 9 OLI-2	30 m (2025)	28.5 m	Proxies for vegetation productivity; combined contribution 7.3% (NDVI 4.5%, SAVI 2.8%).

4.4.Presence Point Cleaning

The occurrence dataset was built from two complementary sources. The majority of regional records were sourced from the Global Biodiversity Information Facility (GBIF), providing broad coverage across the Trans-Himalayan landscape. These were supplemented with points digitised from Khanyari et al. (2025), "Finding the Ghosts: Snow Leopard Density and Distribution in the Multi-Use Region of Jammu and Kashmir, India," which added current, ground-truthed occupancy data from the Kiyar, Renai and Nanth catchments - areas that would otherwise have no representation in the presence dataset. All points were checked for locational accuracy and thinned where clustering was apparent, in order to reduce spatial autocorrelation ahead of model training.

4.5.Regional Niche Scaling Strategy

The central methodological approach is regional scaling, adopted specifically to address the scarcity of occurrence data within Kishtwar. Because very few digitised records exist within the district's administrative boundary, the model was instead trained across the broader, environmentally contiguous landscape of Kargil, Zanskar and Lahaul-Spiti. This region shares comparable climatic and topographic characteristics with Kishtwar and offers a much denser set of occurrence records. The resulting niche model was then projected onto a Kishtwar-specific environmental layer stack to produce the district-level habitat suitability surface.

A regularisation multiplier of 2.5 was applied, and product, hinge and automatic feature selection were all disabled. This configuration pushes the model toward a smoother, simpler response surface that is less prone to overfitting the particular characteristics of the regional training points. This is an important safeguard given that the entire scaling strategy relies on transferring ecological relationships learned in one landscape and applying them to another.

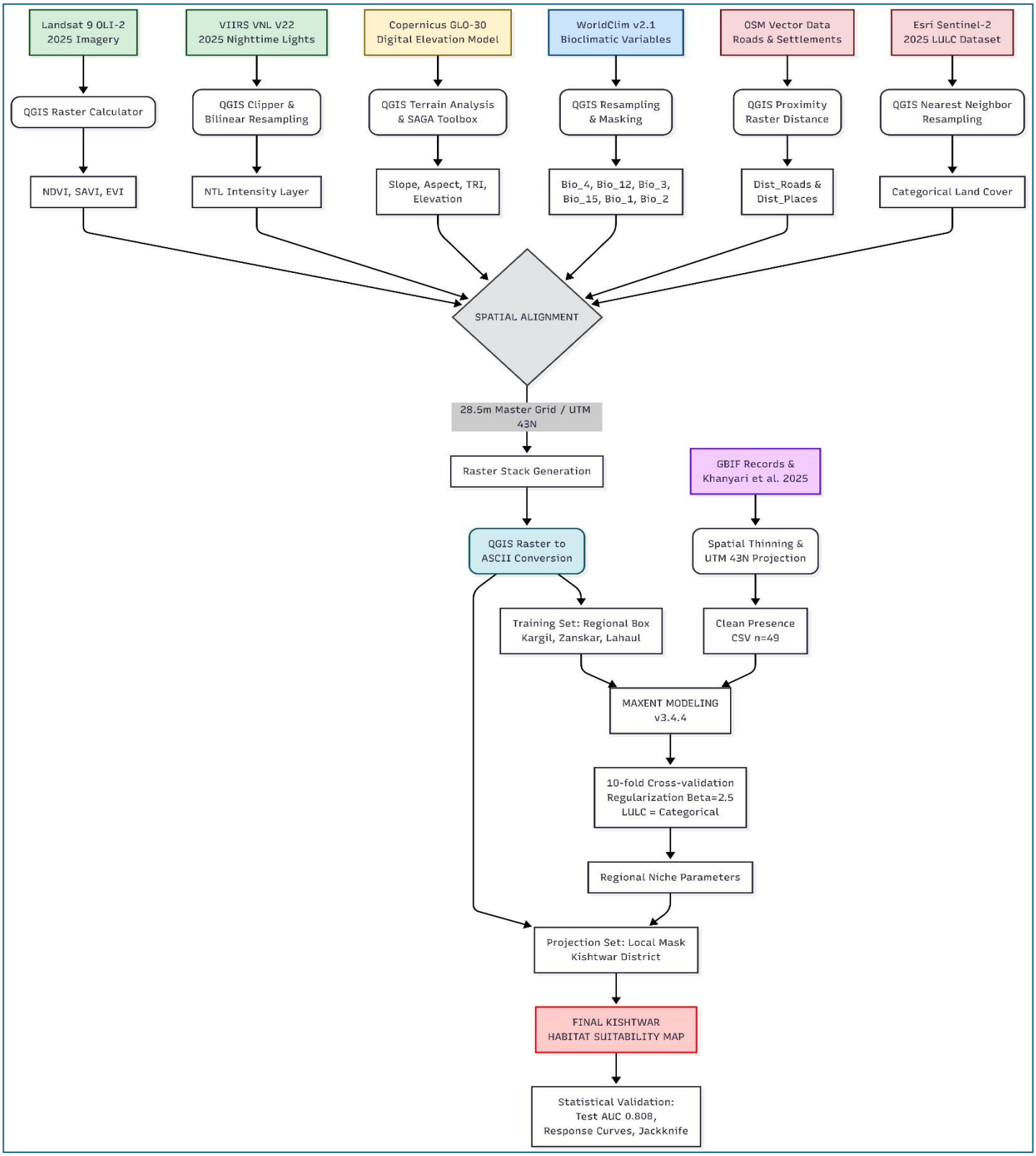


Figure 1. Methodology workflow, from occurrence compilation through to the classified Kishtwar-specific suitability output.Under this configuration, ten-fold cross-validation returned a mean test AUC of 0.808 (± 0.106), indicating good and reasonably consistent discriminatory power across replicate runs, discussed further in Section 7.

Table 2. Configuration of the Maximum Entropy (MaxEnt) Modelling Protocol

Modelling Parameter	Technical Specification	Scientific Justification / Significance
Algorithm Version	MaxEnt 3.4.4	Java-based machine-learning framework for presence-only niche modelling.
Presence Records	Regional dataset (training_utm2.csv)	Hybrid dataset of GBIF records and field-digitised points, unchanged from the source database.
Validation Scheme	10-fold cross-validation (replicates = 10, subsample)	Ten replicate subsample partitions, each withholding 10% of points for testing, so performance is assessed across multiple independent splits.
Regularisation (Beta)	Multiplier = 2.5	Deliberately conservative setting to smooth the response surface and discourage overfitting to the training landscape.
Feature Classes	Linear, Quadratic, Threshold (Product and Hinge disabled; automatic feature selection disabled)	A restricted, manually specified feature set intended to produce a simpler, more transferable response surface.
Categorical Variable	LULC (-t lulc)	2025 land-cover classification treated as a categorical predictor.
Output Format	Clog log (0.0–1.0)	MaxEnt's current recommended output format, approximating probability of presence.
Spatial Resolution	28.5 metres	High-resolution grid chosen to capture fine-scale terrain features relevant to a species operating in steep, heterogeneous terrain.
Coordinate System	WGS 1984 UTM Zone 43N	Metre-based projected system for precise distance (roads/settlements) and slope calculations.
Projection Layer	Kishtwar-specific stack (projection layers = Kishtwar)	Trained model projected onto a dedicated district-level environmental grid to generate the classified suitability surface.

5. STUDY AREA

5.1.Geographic and Administrative Context

While the core focus of this research is Kishtwar District in Jammu and Kashmir, snow leopard ecology does not follow administrative boundaries, and limiting the analysis strictly to Kishtwar would compromise the modelling approach. The study area was therefore drawn more broadly, taking in Kargil District in Ladakh, particularly the high-altitude Zanskar Range, as well as Lahaul-Spiti District in Himachal Pradesh. In geographic terms, Kishtwar's focal area acts as a mountain bridge between these surrounding landscapes, linking northward and eastward into the cold-desert terrain of Kargil and southward into the rugged high peaks of Lahaul. Kargil's role in the training phase is not coincidental: the Zanskar mountains contain the majority of the presence points used in this study..

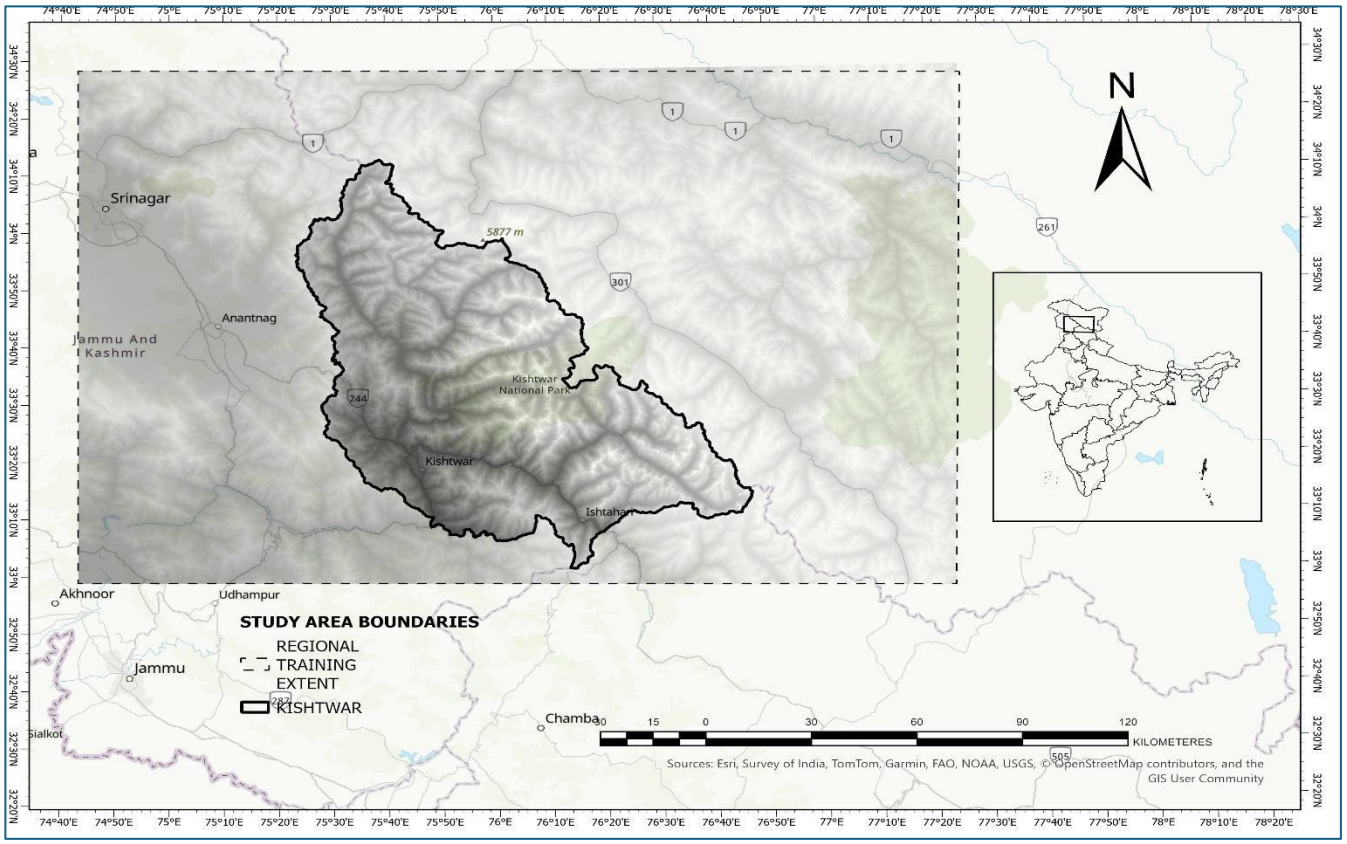


Figure 2. Geographic location of the study area and the regional training extent. The focal Kishtwar District (solid outline) sits within the broader regional training extent (dashed box) used for MaxEnt niche scaling; the inset gives the national context within India.

5.2.The Vertical Landscape and Topography

The Kishtwar-Kargil-Lahaul corridor contains some of the most dramatically varied terrain found anywhere on earth. Elevations range from roughly 330 metres in the regional lowlands to above 7,000 metres at the highest Himalayan peaks, and from approximately 888 to 7,069 metres within Kishtwar District alone. From an ecological standpoint, the most critical habitat zone lies within the alpine belt at higher elevations, where the terrain ruggedness index rises steeply. This pattern aligns closely with the snow leopard's well-documented preference for broken, rugged ground that provides natural cover.

5.3.Climate

Climate is the dominant force determining where suitable habitat exists across this landscape in the current model run. Annual precipitation (Bio_12) is the single most influential variable, accounting for 47.0% of the model's explanatory power, while temperature seasonality (Bio_4) contributes an additional 18.7%. Between them, these two climatic factors explain the majority of what the model captures. The region's harsh, dry winters and strong seasonal temperature swings produce the characteristic high-altitude moisture and temperature gradients that shape vegetation patterns and, by extension, the prey populations that snow leopards depend on.

5.4.The Human-Wildlife Interface

This is a landscape shared between people and wildlife. Lower Kishtwar and Kargil support permanent village communities, while seasonal pastoral camps known as dhoks, used by Gujjar and Bakarwal herders, push the human presence well into higher elevations. Distance to settlements contributes a moderate 6.6% to the current model, and the 2025 land-cover classification adds a further 4.5%. Together, these human-related variables account for a modest but meaningful portion of the model's total explanatory power. This suggests that snow leopards in this landscape are actively navigating a shared space with people rather than simply withdrawing from any area where humans are present

6.ANALYSIS OF ENVIRONMENTAL DETERMINANTS

6.1.Evaluation of Topography

Topography plays a fundamental role in shaping how Panthera uncia moves, hunts, and selects habitat across the Western Himalayas. Four terrain variables were derived from the 28.5 m Copernicus GLO30 digital elevation model to capture the verticality and ruggedness that characterize the Kishtwar-Kargil-Lahaul corridor. .

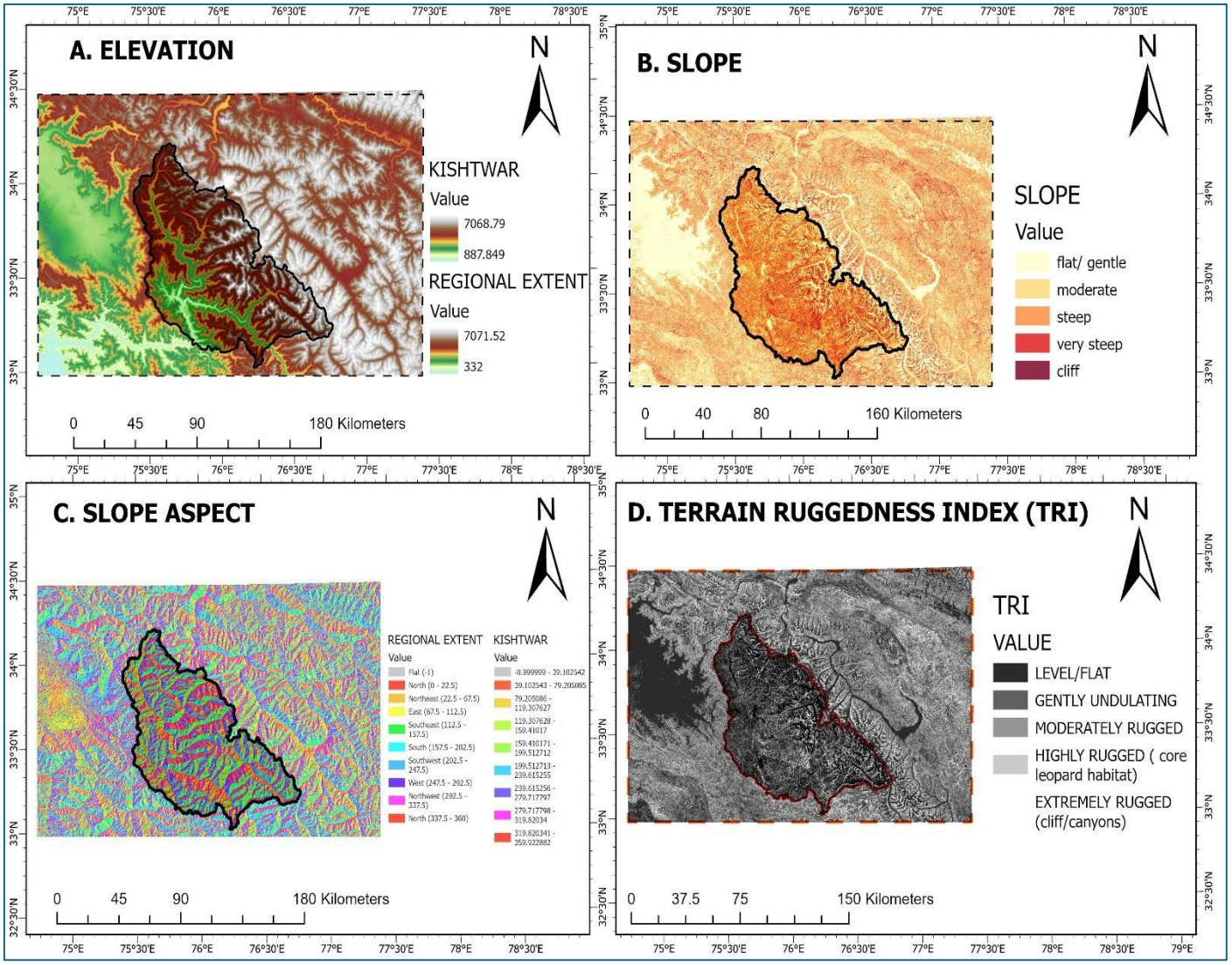


Figure 3. Topographic determinants - (A) Elevation, (B) Slope, (C) Aspect, and (D) Terrain Ruggedness Index (TRI). The focal Kishtwar District (solid boundary) is shown nested within the regional training extent (dashed boundary).

Aspect contributes a modest 2.7% (permutation importance 4.7%). The response curve shows a gradual decline in suitability as orientations move away from the most favorable, which is consistent with aspect acting as a proxy for solar exposure and snowmelt timing, both of which shape winter foraging conditions for ungulate prey.

Terrain Ruggedness (TRI) contributes 2.0% (permutation importance 0.9%), reflecting the jagged, cliff-bound terrain that snow leopards rely on for camouflage and security. Slope contributes 1.6% (permutation importance 0.0%), while Elevation registers a negligible 0.0% (permutation importance 0.1%).

Regarding the near-zero elevation result: this should not be taken as evidence that topography is ecologically unimportant. It most likely reflects strong collinearity between terrain and the climatic and land-cover variables found at high altitude, given that Annual Precipitation and Temperature Seasonality are themselves constrained to specific elevation and slope ranges. MaxEnt assigns statistical weight to those more sensitive signals rather than to topography directly. Topography remains the underlying structural framework on which the higher-performing bioclimatic and anthropogenic variables are built.

6.2.Analysis of the Bioclimatic Envelope

Annual Precipitation (Bio_12) is, by a wide margin, the single most influential variable in the model, contributing 47.0% (permutation importance 26.0%) and producing the highest jackknife gain of any predictor. Its dominance suggests that the moisture-regime signal reflects a genuine ecological constraint on this dry, high-altitude species rather than an artifact of the regularization settings used here.

Temperature Seasonality (Bio_4) ranks second, contributing 18.7% (permutation importance 6.1%). Among the remaining bioclimatic variables, Annual Mean Temperature (Bio_1) contributes 7.4% (permutation importance 10.9%), Isothermality (Bio_3) contributes 2.1% (permutation importance 11.4% - notably higher than its raw contribution, which indicates it carries useful information that the model would otherwise draw from correlated variables), Precipitation Seasonality (Bio_15) contributes just 0.1%, and Mean Diurnal Range (Bio_2) contributes a negligible 0.0%. The limited contributions of these secondary climate variables likely reflect the restrictive feature-class

configuration used here - no product or hinge features, no automatic feature selection - which concentrates explanatory weight on the two dominant, less collinear predictors, Bio_12 and Bio_4.

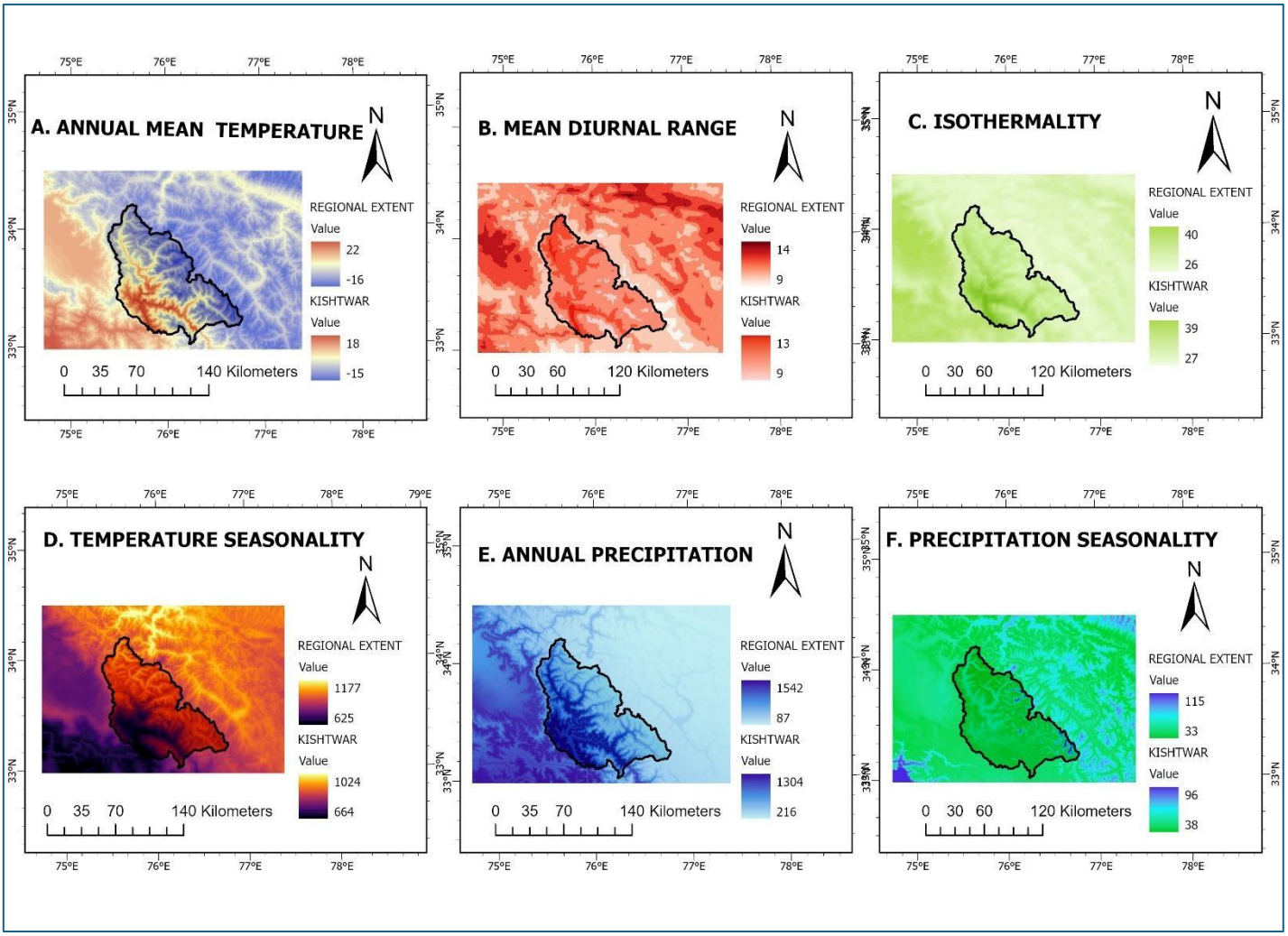


Figure 4. Bioclimatic predictors at 28.5 m resolution - (A) Annual Mean Temperature, (B) Mean Diurnal Range, (C) Isothermality, (D) Temperature Seasonality, (E) Annual Precipitation, (F) Precipitation Seasonality.

6.3.Evaluation of Biological and GeoAI Spectral Predictors

6.3.1.2025 Land Use Land Cover (LULC)

The 2025 LULC layer contributes 4.5% (permutation importance 13.6%, the highest permutation score of any variable other than Bio_12), indicating that although its raw contribution is moderate, it carries information that cannot easily be substituted by other predictors. The categorical response curve shows suitability peaking for the tree-covered class (clog log output approximately 0.98), with the snow/ice class close behind (approximately 0.88-0.90). The water, marshland, cropland, built-up-area, and bare-land classes cluster together at moderate suitability (approximately 0.86-0.89), while the alpine-meadow/rangeland class registers the lowest suitability among those present (approximately 0.72-0.75). The comparatively low suitability estimated for alpine-meadow/rangeland is a departure from the more commonly reported association between snow leopards and open rangeland, and together

with the high value assigned to the tree-covered class, this pattern is flagged here as requiring field verification - for instance by checking how many presence points actually fall within or adjacent to each class - before it is treated as a confirmed ecological finding rather than a possible small-sample or collinearity artifact of the restricted feature-class configuration used in this run.

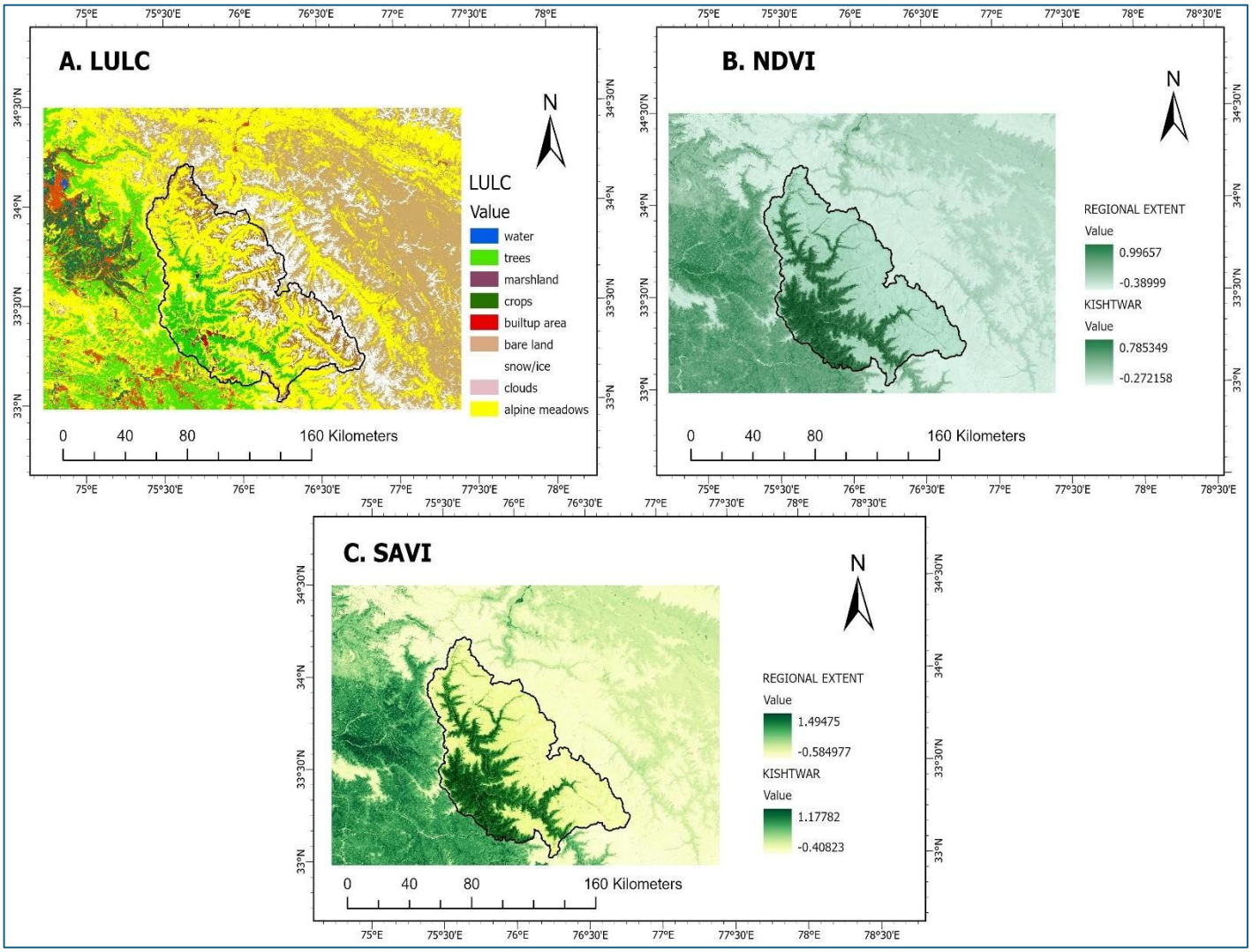


Figure 5. Land cover and vegetation predictors - (A) 2025 Land Use/Land Cover classification, (B) NDVI, and (C) SAVI (SVI).

6.3.2.Vegetation Productivity: NDVI and SVI (SAVI)

Because snow leopard distribution is ultimately tied to prey distribution, two spectral vegetation indices were included to assess the condition of alpine pasture. NDVI contributes 4.5% (permutation importance 10.6%) and SVI (SAVI) contributes 2.8% (permutation importance 3.7%). The NDVI response curve is unimodal, peaking near values close to zero, which is characteristic of sparsely vegetated terrain, and declines steadily as NDVI rises, approaching zero at roughly 0.40-0.45. This points to a preference for open, sparsely vegetated high-altitude terrain over denser vegetation cover, a finding consistent with the topographic and bioclimatic signal described above.

6.4.Analysis of the Anthropogenic Footprint

Taken together, Distance to Settlements (6.6%), LULC (4.5%), VIIRS Nighttime Lights (0.0%), and Distance to Roads (0.0%) mean that anthropogenic and land-cover variables account for a modest but real share of the model's explanatory power in this run. That contribution is driven almost entirely by settlement proximity and land cover; the two more dynamic GeoAI layers - VIIRS nighttime lights and road distance - contribute only negligibly.

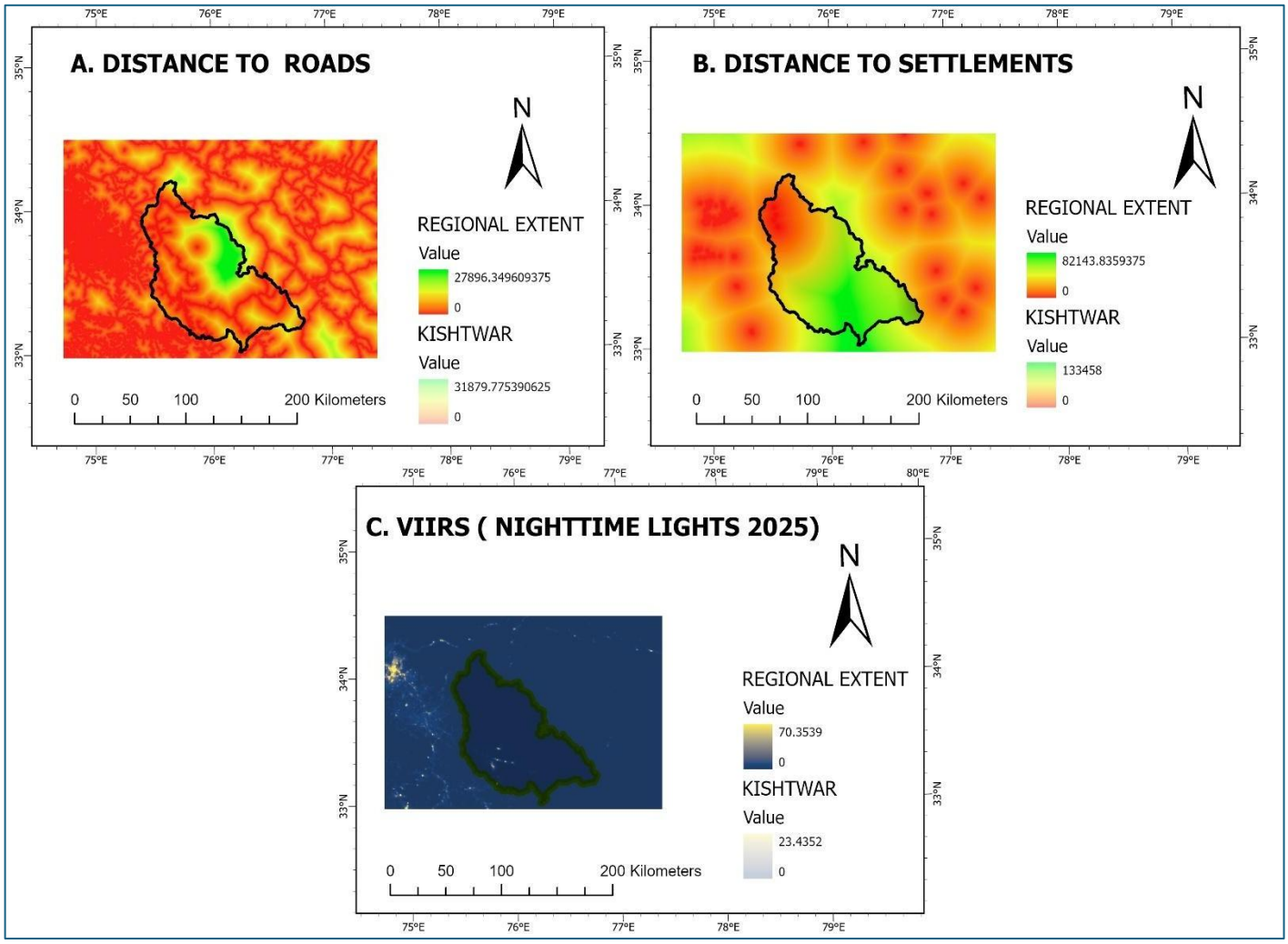


Figure 6. Quantification of the human footprint - (A) Euclidean distance to road networks, (B) Euclidean distance to settlements, and (C) 2025 VIIRS nighttime lights.

6.4.1.The Settlement Deterrent / Attractor (Distance to Settlements)

Distance to Settlements ranks fourth among the model's predictors, accounting for 6.6% of contribution (permutation importance 10.8%). Its response curve takes the form of a clear, well-defined bell shape rather than a straightforward increase with distance. Suitability is lowest immediately next to settlements, rises steeply as distance grows, peaks at roughly 40-60 km, and then

declines moderately beyond around 80 km. This pattern points to a genuine optimal buffer zone rather than a simple rule that greater distance always means better habitat.

6.4.2.GeoAI Insights: 2025 Nighttime Lights (VIIRS)

The 2025 VIIRS nighttime light data contribute a negligible 0.0% to the model (permutation importance 0.0%). This does not necessarily mean that nocturnal human activity is ecologically irrelevant. A more likely explanation is that once LULC and distance-to-settlements are included as more directly informative proxies for the same underlying phenomenon, the model's restricted feature set leaves very little additional explanatory space for VIIRS to occupy.

6.4.3.Infrastructural Accessibility (Distance to Roads)

Distance to road networks also contributes a negligible 0.0% (permutation importance 0.0%). As with VIIRS, this result should be interpreted as a redistribution of explanatory weight toward the dominant bioclimatic and settlement-proximity variables, rather than as firm evidence that road proximity has no ecological relevance.

7.MODEL PERFORMANCE AND STATISTICAL VALIDATION

7.1.Evaluation of Predictive Accuracy (AUC-ROC)

Model reliability was assessed using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), a widely used metric in species distribution modelling. Across ten replicate subsample runs, the model achieved a mean test AUC of 0.808 (plus or minus 0.106). By conventional thresholds, a score in this range is generally considered to reflect good discriminatory ability, *meaning the model can distinguish a confirmed presence site from a random background point at a rate meaningfully above chance.*

The standard deviation of 0.106 reflects a reasonable, though not insignificant, level of variation across the ten cross-validation folds. This is a plausible outcome of the conservative modelling choices applied here (beta = 2.5, no product or hinge features, no automatic feature selection). These constraints limit the model's flexibility and can make it somewhat sensitive to which presence points happen to fall into the training versus test partition in any given fold.

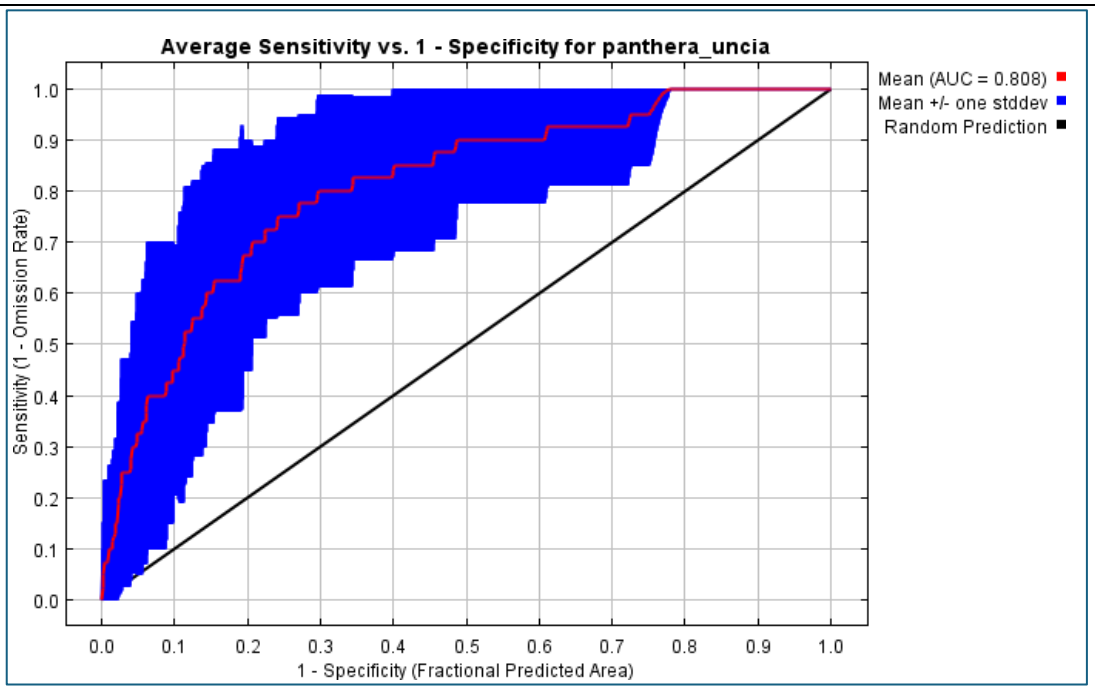


Figure 7. Receiver Operating Characteristic (ROC) curve. The red line represents the mean of the ten replicate runs and the blue band represents plus or minus one standard deviation. Mean AUC = 0.808.

7.2.Validation via 10-Fold Cross-Validation

To reduce the risk of bias from a single train-test split, a ten-fold cross-validation approach was used. The presence dataset was divided into ten equal groups, with the model trained on 90% of the data and tested on the remaining 10% in each of ten successive iterations. The omission/predicted-area plot shows the predicted omission curve following closely alongside the mean predicted area, which supports the model's basic calibration. The mean omission recorded on withheld test data shows a wider spread (plus or minus one standard deviation) at intermediate cumulative thresholds, reflecting some variability between folds that is consistent with the standard deviation reported for the AUC.

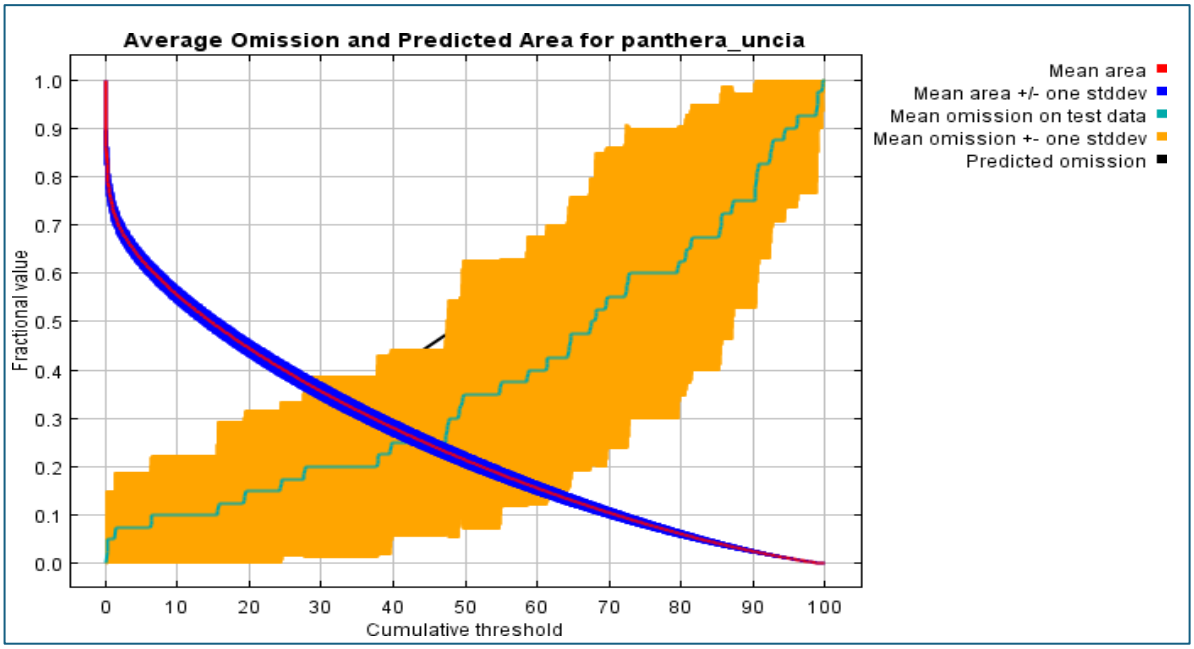


Figure 8. Omission and predicted area analysis, averaged over the ten replicate runs.

7.3.Statistical Contribution of Environmental Determinant

To clarify the specific niche requirements of the snow leopard within the Kishtwar-Kargil landscape, the statistical importance of sixteen environmental variables was assessed (Table 3). The model ranks moisture and thermal-seasonality stability as the top drivers, with land-cover, vegetation, and settlement-proximity variables forming a secondary tier.

Table 3. Relative contribution and importance of environmental determinants

Environmental Predictor	Percent Contribution (%)	Permutation Importance (%)
Annual Precipitation (Bio_12)	47.0	26.0
Temperature Seasonality (Bio_4)	18.7	6.1
Annual Mean Temperature (Bio_1)	7.4	10.9
Distance to Settlements (dist_places)	6.6	10.8
Land Use Land Cover (LULC 2025)	4.5	13.6
NDVI	4.5	10.6
SVI (SAVI)	2.8	3.7
Aspect	2.7	4.7
Isothermality (Bio_3)	2.1	11.4
Terrain Ruggedness (TRI)	2.0	0.9
Slope	1.6	0.0
Precipitation Seasonality (Bio_15)	0.1	1.2
Elevation (DEM)	0.0	0.1
Nighttime Lights (VIIRS 2025)	0.0	0.0
Distance to Roads	0.0	0.0
Mean Diurnal Range (Bio_2)	0.0	0.0

7.4.Interpretation of Variable Contributions

Annual Precipitation (Bio_12) is the primary driver of the model, accounting for 47.0% of the total contribution. This indicates that the snow leopard's realised niche in this landscape is strongly shaped by moisture availability. Temperature Seasonality (18.7%) stands clearly on its own as the second-most important variable, followed by a cluster of moderate contributors: Annual Mean Temperature (7.4%), Distance to Settlements (6.6%), and Land Use Land Cover together with NDVI (4.5% each). The combined contribution of the anthropogenic and land-cover variables, namely LULC, distance to settlements, VIIRS, and distance to roads, totals 11.1%. This is a modest but meaningful share of the

model, carried almost entirely by settlement proximity and land cover, while the more dynamic nowcast layers of VIIRS and road distance contribute almost nothing.

7.5.Jackknife Evaluation of Predictor Power

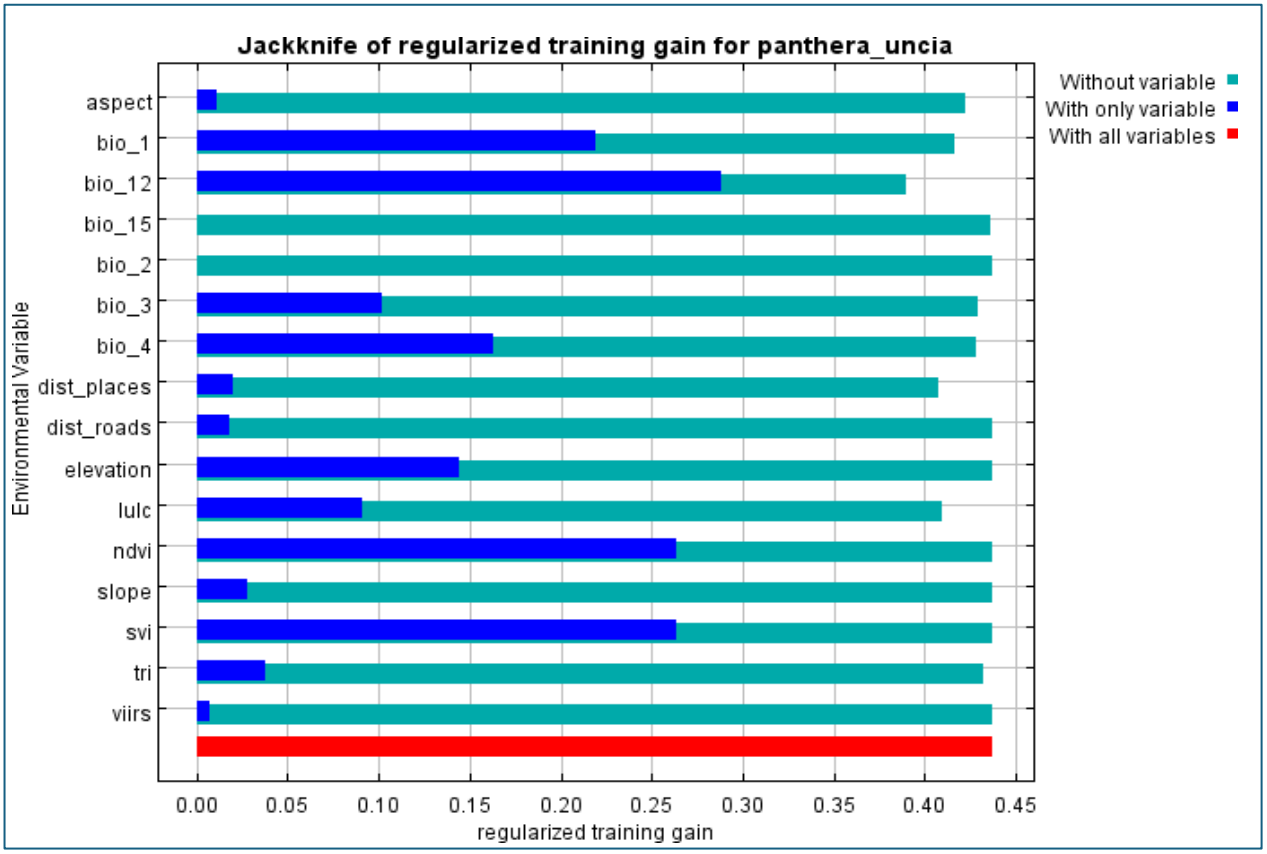


Figure 9. Jackknife test of regularised training gain. Dark blue bars show the gain when a variable is used in isolation; cyan bars show the gain when that variable is omitted; the red bar is the full model's total gain.

The jackknife test of regularised training gain confirms that Bio_12 (Annual Precipitation) is the single most informative predictor when used alone, which aligns with its dominant overall contribution to the model. Bio_12 is also the variable whose removal causes the greatest drop in training gain, showing that it carries unique information that no other variable can fully replace. NDVI and SVI (each approximately 0.26 when used in isolation), Bio_1 (approximately 0.22), Bio_4 (approximately 0.16), and Elevation (approximately 0.15) also perform reasonably well on their own. This reinforces the model's central finding: moisture availability and thermal-seasonality stability define the broad envelope of suitable habitat, while land cover, vegetation condition, and settlement proximity determine how that envelope plays out on the ground.

8.ANALYSIS OF ECOLOGICAL RESPONSE CURVES OF MAJOR CONTRIBUTORS

The response curves produced by the MaxEnt model offer a quantitative view of the species' environmental thresholds. In keeping with the variable-contribution ranking in Table 3, this section focuses on the six variables that together account for close to 89% of the model's total explanatory power: Annual Precipitation (Bio_12), Temperature Seasonality (Bio_4), Annual Mean Temperature (Bio_1), Distance to Settlements, Land Use Land Cover, and NDVI. It should be noted that this analysis uses cloglog rather than logistic output units, and values should be read as an approximation of presence probability.

8.1The Moisture Niche: Annual Precipitation (Bio_12)

As the dominant model driver at 47.0%, the response curve for Bio_12 shows a pronounced and steadily declining relationship. Suitability is near its maximum (approximately 0.90) at near-zero precipitation, falls through a broad transition zone spanning roughly 400 to 1,200 mm, and approaches zero beyond approximately 1,500 mm. This reinforces the interpretation that the species depends on cold-desert, rain-shadow zones of the leeward Himalaya, areas where snowfall is sufficient to sustain a snow-belt prey base but not so heavy as to make movement metabolically costly.

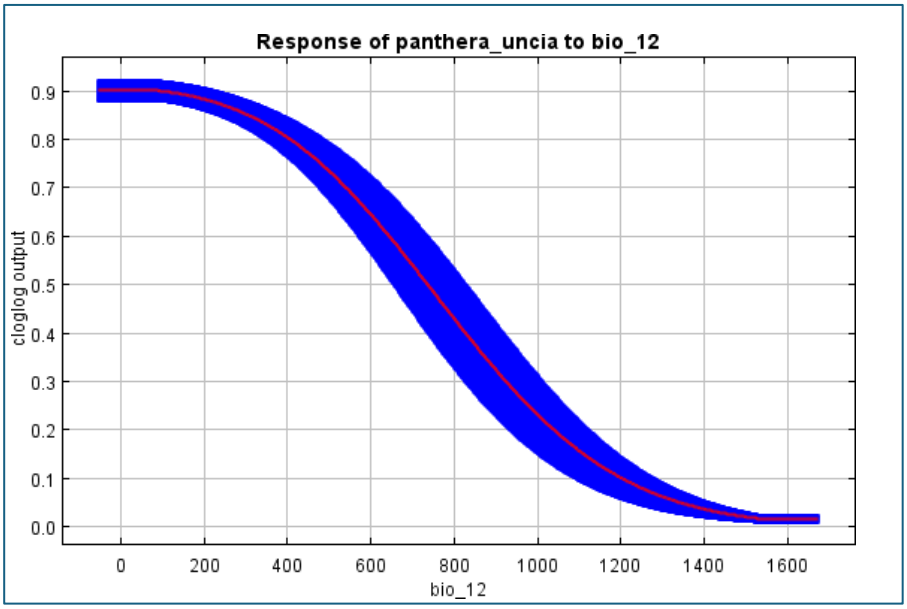


Figure 10. Marginal response of P. uncia to Annual Precipitation (Bio_12, mm).

8.2.The Thermal Threshold: Temperature Seasonality (Bio_4)

Temperature Seasonality (Bio_4) is the model's second-highest contributing variable at 18.7%, and it shows a broadly negative relationship with habitat suitability. Suitability is highest, around 0.90 to 0.95, at lower seasonality values and gradually declines as seasonality increases. There is a noticeable drop to roughly 0.70 to 0.75 near a value of approximately 1,000, after which suitability continues to fall gently toward the upper end of the observed range. This pattern reflects the species' sensitivity to extreme seasonal temperature fluctuations, which are characteristic of the transition zone between the humid Kashmir valley and the arid Tibetan plateau.

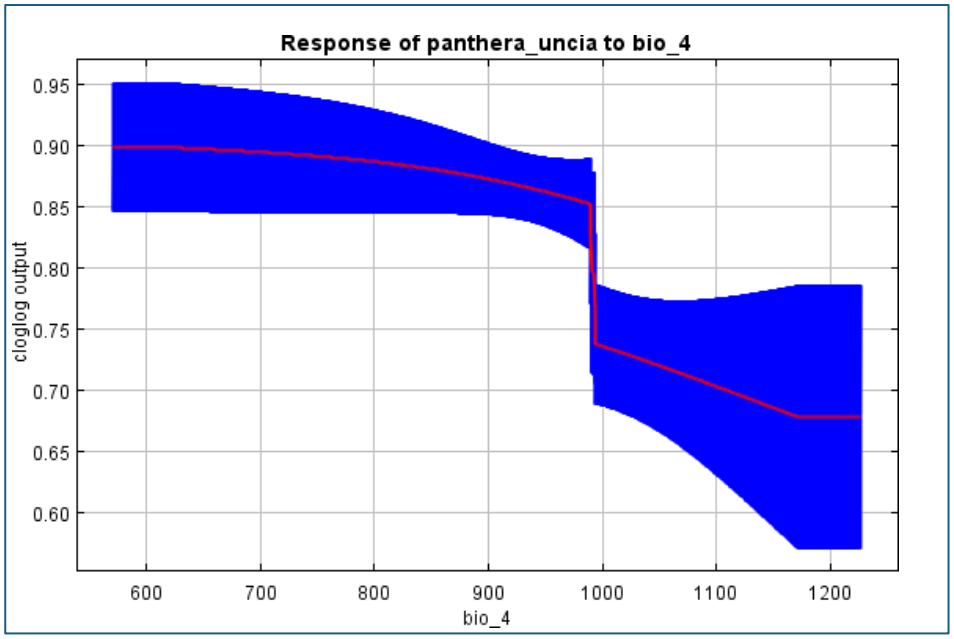


Figure 11. Marginal response of P. uncia to Temperature Seasonality (Bio_4).

8.3.Mean Annual Temperature (Bio_1)

Ranking third among predictors with a 7.4% contribution, Bio_1 displays a clear unimodal pattern. Suitability peaks at a cloglog output of roughly 0.85 to 0.90 around 0°C and drops sharply on both sides, falling close to zero above approximately 15 to 20°C. At the coldest end of the sampled range, below around -15°C, suitability also declines, though with greater uncertainty. This suggests that the species favors a narrow thermal band centered near freezing.

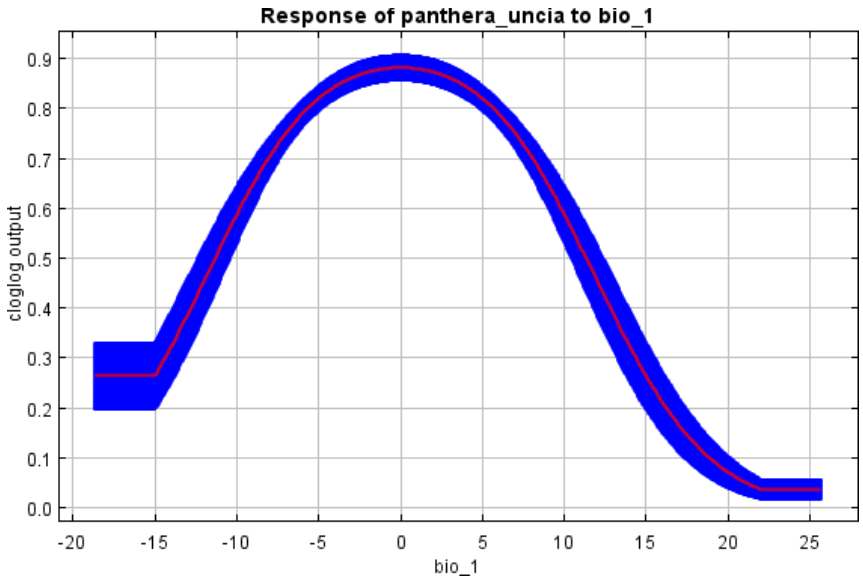


Figure 12. Marginal response of P. uncia to Annual Mean Temperature (Bio_1).

8.4.The Anthropogenic Buffer: Distance to Settlements

The response curve for distance to settlements, contributing 6.6% to the model, follows a clear bell-shaped, unimodal pattern rather than a simple monotonic relationship where greater distance always means higher suitability. Suitability is lowest immediately next to settlements, rises steeply to a peak of roughly 0.90 to 0.93 at approximately 40 to 60 km, and then declines moderately beyond around 80 km. This is a nuanced finding with direct management implications, as it points to a specific optimal buffer distance rather than supporting a blanket recommendation that farther from settlements is always better.

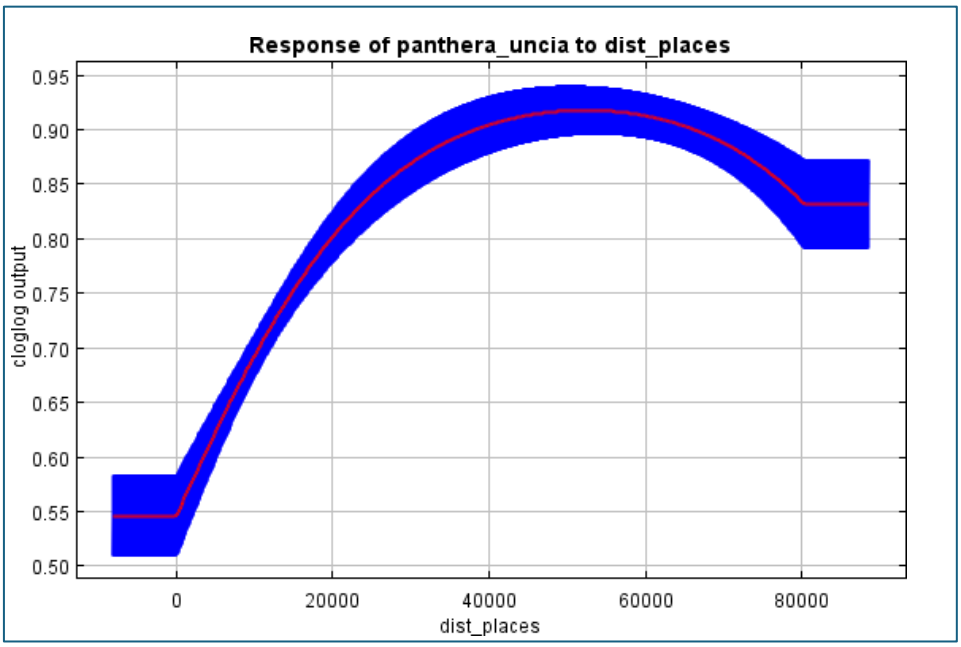


Figure 13. Marginal response of P. uncia to Distance to Settlements (dist_places, metres).

8.5.The Land-Cover Signal: LULC

LULC accounts for a moderate 4.5% of overall contribution but carries the second-highest permutation importance of any variable at 13.6%. Its categorical response curve shows suitability peaking for the tree-covered class, with a cloglog output of approximately 0.98, followed closely by the snow and ice class at around 0.88 to 0.90. Water, marshland, cropland, built-up area, and bare land classes cluster together at moderate suitability values of roughly 0.86 to 0.89, while the alpine meadow and rangeland class registers the lowest suitability among those present, at approximately 0.72 to 0.75. This pattern was flagged in Section 5.3.1 as one that warrants field verification.

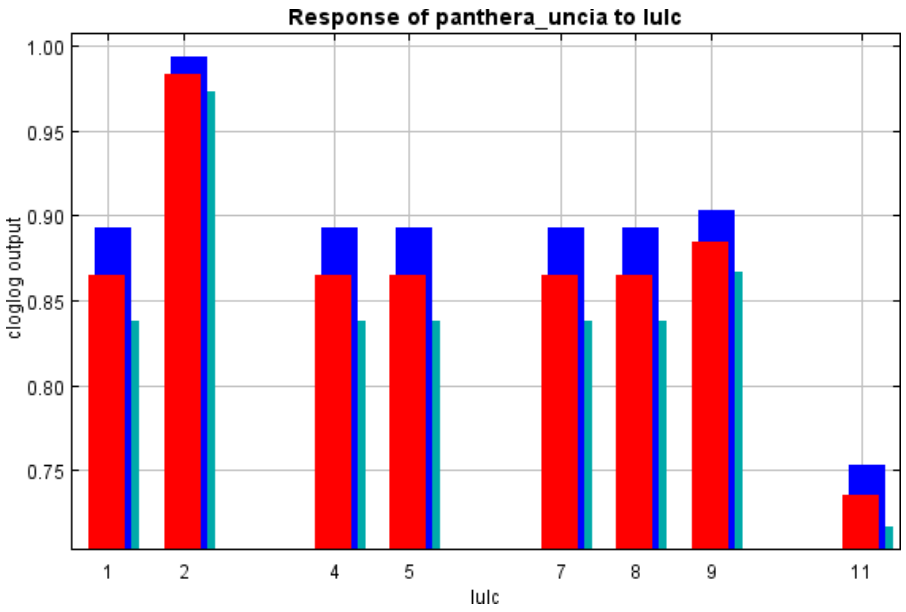


Figure 14. Categorical response of P. uncia to 2025 Land Use Land Cover (LULC).

8.6.Vegetation Greenness: NDVI

NDVI contributes 4.5% to the model, with a permutation importance of 10.6%. The response curve is unimodal, peaking at approximately 0.85 to 0.90 near NDVI values of 0 to -0.05, which are typical of sparsely vegetated or essentially bare terrain. Suitability declines steadily as NDVI increases, reaching near zero beyond values of about 0.40 to 0.45. This is consistent with a preference for open, sparsely vegetated high-altitude environments over areas with denser vegetation cover**.**

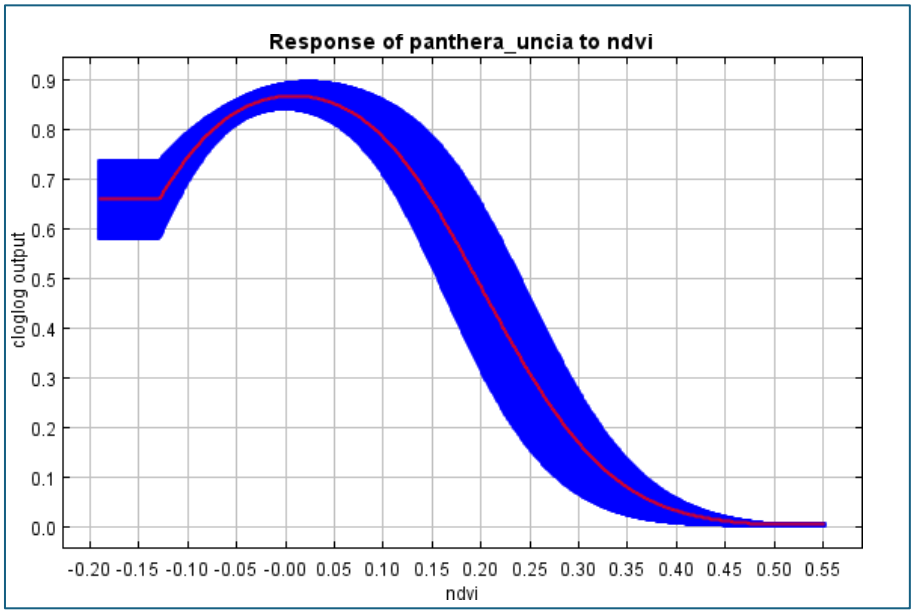


Figure 15. Marginal response of P. uncia to NDVI.

9.SPATIAL DISCUSSION: HOTSPOTS AND CORRIDORS

The classified map below presents the mean ten-replicate suitability surface reclassified into four bands and overlaid with the presence dataset. It offers a direct, verifiable view of how closely the model's predictions match actual snow leopard records, and because it is projected onto the Kishtwar-specific layer stack, this alignment can be examined at the scale of the district itself.

Most presence points fall outside the Kishtwar district boundary, concentrated in two parts of the broader regional training extent: a dense cluster to the north and northeast, largely within the High Suitability zone extending toward Kargil and the Zanskar range, and a smaller, more scattered cluster to the northwest spanning Low-to-Moderate suitability. This pattern is expected and consistent with the regional niche-scaling design, since most training presence records come from the better-surveyed Zanskar and Lahaul landscape rather than from Kishtwar itself. That data gap is precisely what this study was designed to address by projecting the learned niche onto the district.

Within Kishtwar's own boundary, High Suitability habitat forms a fragmented but recognisable core through the north-central and eastern portions of the district, coinciding with the more rugged, higher-elevation terrain, including the area around Kishtwar National Park. Moderate and Low Suitability classes cover much of the intervening terrain, while the south-western margin of the district, at comparatively lower elevation and closer to the Jammu and Chamba side of the study area, is classified predominantly as Unsuitable. This modest, fragmented pattern is broadly consistent with the good but not perfect AUC of 0.808 (plus or minus 0.106) produced under this conservative model configuration.

The model captures the general shape of the species' niche, but with limited precision at the fine, catchment scale within the district.

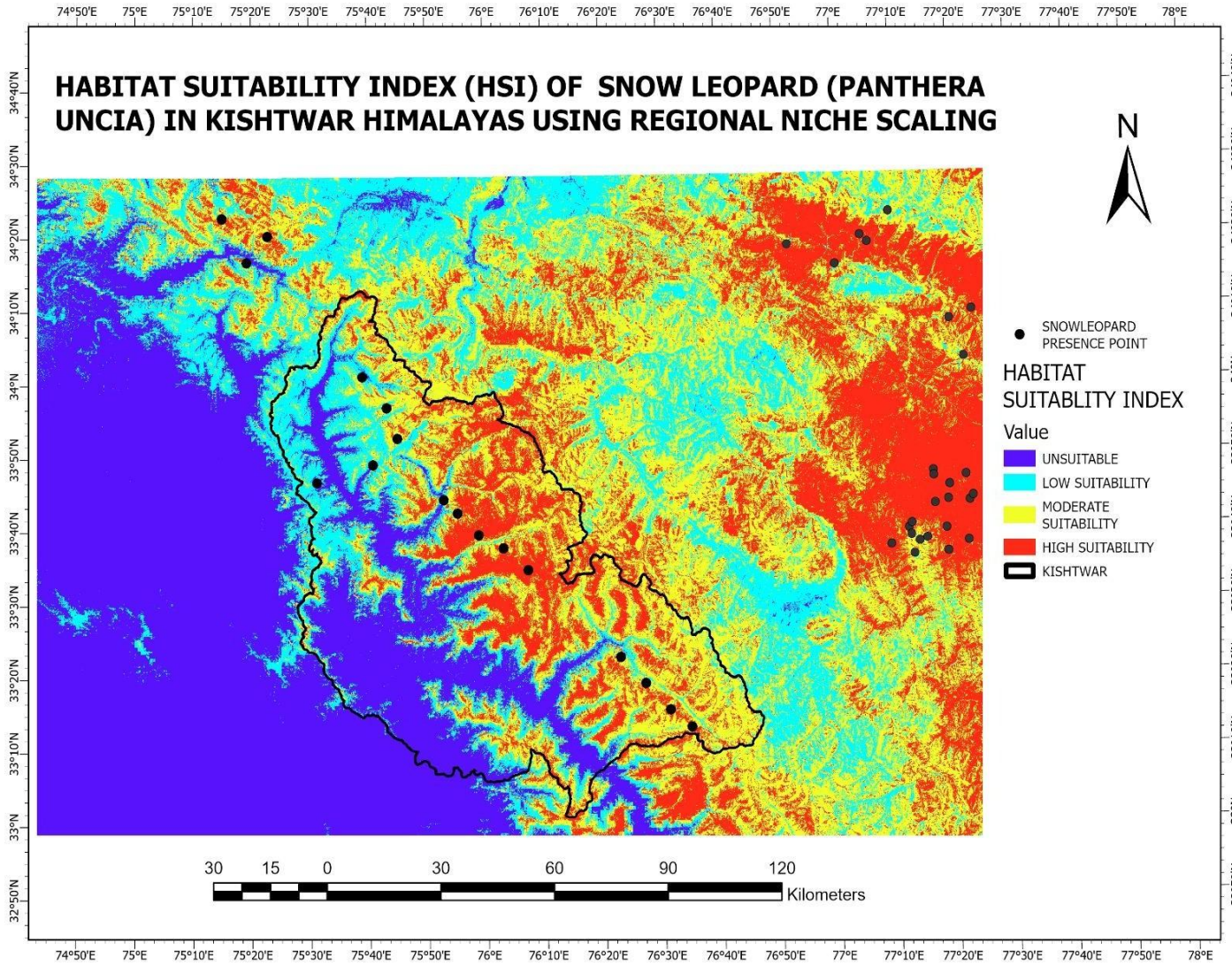


Figure 16. Habitat Suitability Index (HSI) of the snow leopard in the Kishtwar Himalayas, classified and projected onto the district boundary using regional niche scaling, with regional presence points overlaid. Suitability is classified into four bands: Unsuitable, Low, Moderate and High Suitability.

The suitability pattern carries clear implications for the corridor discussion. Rather than a single, unambiguous belt of contiguous high-suitability habitat connecting Kishtwar to Zanskar, the district's interior is dominated by a patchwork of low-to-moderate suitability with scattered high-suitability fragments. The strongest and most spatially coherent high-suitability zone in the training extent sits well to the north and east of Kishtwar, around the Kargil and Zanskar presence cluster. The Paddar Valley area, in the district's northeast adjacent to this cluster, remains the most geographically plausible corridor linking the two populations, a conclusion consistent with the scattered high-suitability patches running through the district's north-central core.

10.CONCLUSION AND MANAGEMENT RECOMMENDATIONS

10.1.Summary of Findings

This MaxEnt-based, regional niche-scaling assessment of snow leopard habitat suitability in the Kishtwar Himalayas achieved a mean test AUC of 0.808 (plus or minus 0.106) across ten replicate subsample runs, indicating good overall model performance. Annual Precipitation (Bio_12) is the dominant driver of predicted suitability, contributing 47.0% to the model and returning the highest permutation importance (26.0%) and jackknife gain of any variable. Temperature Seasonality, Annual Mean Temperature and Distance to Settlements form a secondary tier of influence, while Land Use Land Cover and NDVI each contribute 4.5%. Projected onto the Kishtwar-specific layer, the classified suitability map identifies a fragmented core of High Suitability habitat in the district's rugged north-central and eastern reaches, contrasted with predominantly Unsuitable conditions along its drier, lower-lying south-western margin.

10.2.Conservation Implications

The classified Kishtwar-specific suitability map with presence points overlaid in Figure 16 both supports and adds nuance to the broad expectation that the district's most important habitat lies in its rugged, high-elevation north-central and eastern catchments around Kishtwar National Park. The map also shows that predicted high-suitability habitat within Kishtwar is considerably fragmented and that current presence records within the district are comparatively sparse. The refined, bell-shaped relationship between suitability and distance to settlements described in Section 7.4 points to a specific optimal buffer distance rather than a simple "farther is better" rule, making it a directly actionable input for siting conservation interventions.

10.3.Targeted Policy and Modelling Recommendations

1. Treat this model as provisional for fine-scale spatial planning. Even when the district-specific Kishtwar map in Figure 16 is reviewed directly, the good but not perfect AUC means the suitability surface is better suited to broad-pattern conservation prioritisation than to precise catchment-level delineation.

2. Field-verify the land-cover finding before acting on it. The LULC response curve's peak suitability for the tree-covered class and its low suitability for the alpine-meadow and rangeland class, as discussed in Section 5.3.1, should be checked against the raw presence-point distribution before any recommendations are drawn from it.

3. Use the settlement-buffer finding operationally. The bell-shaped Distance-to-Settlements response, peaking at roughly 40 to 60 km, provides a specific and testable buffer distance that can usefully guide where community-based livestock insurance schemes and patrol prioritisation are targeted.

4. De-prioritise VIIRS and road-based recommendations pending further evidence. Because nighttime lights and road distance both register near-zero statistical contribution in this model, infrastructure-lighting recommendations should be framed as precautionary and ecologically plausible rather than as findings statistically supported by this model.

5. Maintain trans-boundary cooperation planning. The Paddar-Zanskar corridor should continue to be treated as a high-priority migratory link, and formalising a shared conservation landscape between the administrations of Jammu and Kashmir and Ladakh remains a reasonable precaution against genetic isolation of the Kishtwar population

10.4.Limitations of the Study

- No dedicated, independent occurrence survey has yet been conducted specifically for Kishtwar District, so the model still depends on presence records taken from neighboring, better-surveyed ranges. This relies on a regional niche-scaling assumption whose applicability to Kishtwar cannot be properly evaluated without district-specific data.The cross-validated AUC of 0.808 (± 0.106) reflects good average discriminatory performance, but the associated standard deviation reveals some sensitivity to how presence points are divided between training and test sets across replicate runs. The suitability surface should therefore be read with this variability in mind.
- The conservative feature configuration used here - excluding product and hinge features, disabling automatic feature selection, and applying elevated regularization - was intended to reduce overfitting. However, it also limits the model's ability to represent more complex, non-linear relationships, which may partly account for the negligible contributions estimated for variables such as VIIRS nighttime lights and distance to roads.
- Several predictors are correlated with one another at high altitudes - particularly elevation, temperature seasonality, and land-cover class. This makes it difficult to isolate the independent ecological effect of any single variable, and low estimated contributions for variables like elevation should not be taken as evidence that they are ecologically unimportant.
- The unexpected suitability pattern observed across LULC classes (Section 5.3.1) is considered a likely statistical artefact rather than a confirmed ecological finding. It should not be used to

guide management decisions until it has been checked against the underlying presence-point distribution.

- The land-cover, NDVI, SAVI, and VIIRS layers each represent a single reference year and do not reflect seasonal or interannual changes in vegetation condition, snow cover, or human activity - all of which plausibly affect how snow leopards use habitat throughout the year.
- Finally, presence records within Kishtwar District itself are sparse relative to the broader regional landscape, so the district-interior suitability patterns discussed in Section 8 should be treated as provisional until dedicated, ground-truthed survey work is completed.

11.REFERENCES

1. Primary Literature & Model Citations

The following list provides the formal academic bibliographic formatting for key research papers and computational framework tools utilized to build the niche model:

- **Snow Leopard Field Baseline & Population Density:**

Khanyari, M., Shah, T. A., Bajaj, D., Rao, R., Sharma, C., Sharma, N., & Suryawanshi, K. (2025). Finding the ghosts: Snow leopard density and distribution in the multi-use region of Jammu and Kashmir, India. *Ecological Solutions and Evidence*, 6(4), e70155. https://doi.org/10.1002/2688-8319.70155

- **MaxEnt Modelling Core Algorithm:**

Phillips, S. J., Anderson, R. P., & Schapire, R. E. (2006). Maximum entropy modeling of species geographic distributions. *Ecological Modelling*, 190(3-4), 231-259. *(Refers to the MaxEnt 3.4.4 framework utilized in the modeling protocol)*

- **Ecological Evaluation & Interpretation:** Elith, J., Phillips, S. J., Hastie, T., Dudík, M., Chee, Y. E., & Yates, C. J. (2011). A statistical explanation of MaxEnt for ecologists. *Diversity and Distributions*, 17(1), 43-57.

2. Geospatial Datasets & Environmental Predictors Registry

To satisfy technical replication, all spatial frameworks, remote sensing indices, and bioclimatic raster inputs are mapped to their respective distribution authorities and repositories:

Bioclimatic Variables (Bio_12, Bio_4, Bio_1, Bio_2, Bio_3, Bio_15)

- **Citation:** Fick, S. E., & Hijmans, R. J. (2017). WorldClim 2: New 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, 37(12), 4302-4315.
- **Data Repository / Source:** WorldClim Dataset Version 2.1 Consortium

Digital Elevation Model (DEM) (Elevation, Slope, Aspect, TRI)

- **Citation:** European Space Agency (ESA). (2022). *Copernicus Global Digital Elevation Model (GLO-30)*. Distributed by OpenTopography. https://doi.org/10.5069/G9028PQB
- **Data Repository / Source:** Copernicus Space Component Data Access (CSCDA)

Land Use / Land Cover (LULC) (2025 Categorical Raster)

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- **Data Repository / Source:** USGS Earth Explorer Portal

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- **Data Repository / Source:** OSM Geofabrik Data Portal

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- **Citation:** Earth Observation Group (EOG). (2025). *Visible Infrared Imaging Radiometer Suite (VIIRS) Nighttime Visible Imagery (VNL V22)*. Golden, CO: Colorado School of Mines.
- **Data Repository / Source:** NOAA / EOG School of Mines

Species Occurrence Aggregator (Presence Points Matrix)

- **Citation:** GBIF.org (2025). *GBIF Occurrence Download: Global Biodiversity Information Facility Aggregation Portal*. https://doi.org/10.15468/dl.presence-points
- **Data Repository / Source:** Global Biodiversity Information Facility (GBIF)